

ASSIGNMENT # 01

Subject: PF



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BS (CS)

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DEPARTMENT OF COMPUTER SCIENCE

RIPHAH INTERNATIONAL UNIVERSITY SAHIWAL CAMPUS

Question #01

Algorithm

Begin

1. Input Values:

- Total fee of a semester.
- Number of subjects in a semester.
- Monthly transport fee.

2. Calculate Total Number of Classes in a Semester:

- Total weeks in a semester = 16
- Classes per week per subject = 3
- Total classes per subject = $16 * 3$
- Total classes for all subjects = Total classes per subject * Number of subjects

3. Calculate Financial Losses:

- Loss for missing one class: Semester fee / Total number of classes
- Loss for missing all classes in a day: Loss for missing one class * Total classes in a day
- Transport loss for one day: Monthly transport fee / 30
- Total loss for missing all classes in a day: Loss for missing all classes + Transport loss
- Loss for being absent for 10 days: (Loss for missing all classes in a day + Transport loss for one day) * 10

4. Output the Results:

- Financial loss for missing one class.
- Financial loss for missing all classes in a day.
- Financial loss for being absent for 10 days.

End

Code:

```
#include<stdio.h>
```

```

int main(){
//input
int sem_fee , transport_fee;
int num_subjects;
printf("total_sem_fee :");
scanf("%d",&sem_fee);
printf("transoport_fee :");
scanf("%d",&transport_fee);
printf("num_subjects :");
scanf("%d",&num_subjects);
//data required
const int weeks_per_sem=16;
const int class_per_sub_per_week=3;
const int total_classes_per_sub_sem=weeks_per_sem*class_per_sub_per_week;
const int total_classes_all_sub_per_sem=total_classes_per_sub_sem*num_subjects;
//calculations
float total_cost=sem_fee+transport_fee;
float cost_one_class=total_cost/total_classes_all_sub_per_sem;
float loss_all_classes_per_day=3*cost_one_class;
float absent_10working_days=loss_all_classes_per_day*10;
printf("The Financial loss of father of a std. who missed one class in a day= %f\n",cost_one_class);
printf("The Financial loss of father of a std. who missed all classes of a day=
%f\n",loss_all_classes_per_day);
printf("The Financial loss of father of a std. who remains absent 10 days in a semester=
%f\n",absent_10working_days);
}

```

ScreenShot:

```
total_sem_fee :107000
transport_fee :30000
num_subjects :7
The Financial loss of father of a std. who missed one class in a day= 407.738098
The Financial loss of father of a std. who missed all classes of a day= 1223.214355
The Financial loss of father of a std. who remains absent 10 days in a semester= 12232.143555

-----
Process exited after 67.05 seconds with return value 0
Press any key to continue . . .
```

Question #02

Code:

```
#include<stdio.h>

int main(){

    //inputs

    int total_citizens , tolls_per_day , carpool_citizens;

    float avg_miles_per_day ,cost_per_litter, avg_miles_per_litter , avg_parking_per_day;

    printf("total citizen of sahiwal who daily driving their own cars :");

    scanf("%d",&total_citizens);

    printf("average miles driven per day by a citizen :");

    scanf("%f",&avg_miles_per_day);

    printf("cost per litter of petrol :");

    scanf("%f",&cost_per_litter);

    printf("average miles per litter :");

    scanf("%f",&avg_miles_per_litter);

    printf("average parking fee per day :");

    scanf("%f",&avg_parking_per_day);

    printf("tolls per day payed by citizen :");

    scanf("%d",&tolls_per_day);

    printf("total num of citizens who will share a car when carpool approach is adopted :");

    scanf("%d",&carpool_citizens);

    //calculations
```

```

float total_petrol=avg_miles_per_day/avg_miles_per_litter;
float total_petrol_cost = total_petrol * cost_per_litter;
float total_cost_per_day_per_citizez=total_petrol_cost+tolls_per_day+avg_parking_per_day;
float total_cost_per_day_total_citizens=total_cost_per_day_per_citizez*total_citizens;
float total_per_day_saving=total_cost_per_day_per_citizez*carpool_citizens;
float total_carbon_emit_per_person=total_petrol*2.31;
float total_carbon_emit_all_citizens=total_carbon_emit_per_person*total_citizens;
float total_carbon_reduced=total_carbon_emit_per_person*carpool_citizens;
//output
printf("total petrol is %.2f \n ",total_petrol);
printf("total petrol cost is %.2f \n ",total_petrol_cost);
printf("total cost per day per citizen is %.2f \n ",total_cost_per_day_per_citizez);
printf("total carbon emitted by all citizens is %.2f \n",total_carbon_emit_all_citizens);
//final outputs
printf("total cost per day total citizens is %.2f \n ",total_cost_per_day_total_citizens);
printf("total per day saving of citizens is %.2f \n",total_per_day_saving);
printf("total carbon reduced is %.2f kg \n",total_carbon_reduced);
}

```

ScreenShot:

The screenshot shows the Dev-C++ IDE with the file `D:\que2.c` open. The code is a C program that calculates various costs and savings for citizens. The code is as follows:

```
1 #include<stdio.h>
2 int main(){
3     //inputs
4     int total_citizens , tolls_per_day , carpool_citizens;
5     float avg_miles_per_day , cost_per_litter , avg_miles_per_litter , ;
6     printf("total citizen of sahiwal who daily driving their own cars :");
7     scanf("%d",&total_citizens);
8     printf("average miles driven per day by a citizen :");
9     scanf("%f",&avg_miles_per_day);
10    printf("cost per litter of petrol :");
11    scanf("%f",&cost_per_litter);
```

The output window shows the following results:

```
D:\>gcc -o que2 que2.c
D:\>que2
total citizen of sahiwal who daily driving their own cars :1000
average miles driven per day by a citizen :50
cost per litter of petrol :270
average miles per litter :12
average parking fee per day :150
tolls per day payed by citizen :200
total num of citizens who will share a car when carpool approach is adopted :30
total petrol is 4.17
total petrol cost is 1125.00
total cost per day per citizen is 1475.00
total carbon emitted by all citizens is 9625.00
total cost per day total citizens is 1475000.00
total per day saving of citizens is 44250.00
total carbon reduced is 288.75 kg
D:\>
```

Question #03

Code:

```
#include<stdio.h>
```

```
int main(){
```

```
    //inputs
```

```
int emp_id , late_arrival , total_leave , allow_leaves, tax_rate;
```

```
char emp_name[100];
```

```
float gross_salary ;
```

```
printf("enter Employee ID :");
```

```
scanf("%d",&emp_id);
```

```
printf("enter Employee Name :");
scanf("%s",&emp_name);
printf("enter Gross salary of the employee :");
scanf("%f",&gross_salary);
printf("enter Income tax rate in percent :");
scanf("%d",&tax_rate);
printf("enter total leave of the employee in a month :");
scanf("%d",&total_leave);
printf("enter Number of Allowed Leaves per Month :");
scanf("%d",&allow_leaves);
printf("enter Total Number of Days the Employee Arrived Late:");
scanf("%d",&late_arrival);

    //calculations
float tot_tax=(tax_rate*gross_salary)/100;

float per_day_salary=gross_salary/30;
float half_day_salary=per_day_salary/2;
//late arrival find
float late_arrival_fine=half_day_salary*late_arrival;
//total leaves find
float extra_leaves=total_leave-allow_leaves;
float extra_leaves_fine=extra_leaves*per_day_salary;
//net salary
float net_salary=gross_salary-tot_tax-late_arrival_fine-extra_leaves_fine;

//output
```

```
printf("Emp id\t emp name\t gross salary\t tax rate\t tot tax\t late arrival\t tot leave\t net salary\n");
```

```
printf("-----\n");
```

```
printf("%d\t %s\t %.2f\t %d\t %.2f\t %d\t %d\t %.2f\n",emp_id,emp_name,gross_salary,tax_rate,tot_tax,late_arrival,total_leave,net_salary);
}
```

ScreenShot:

The screenshot shows a C++ IDE with the following code in `que3.c`:

```

8  scanf("%d",&emp_id);
9  printf("enter Employee Name :");
10 scanf("%s",&emp_name);
11 printf("enter Gross salary of the employee :");
12 scanf("%f",&gross_salary);
13 printf("enter Income tax rate in percent :");
14 scanf("%d",&tax_rate);
15 printf("enter total leave of the employee in a month :");
16 scanf("%d",&total_leave);
17 printf("enter Number of Allowed Leaves per Month :");
18 scanf("%d",&allow_leaves);
19 printf("enter Total Number of Days the Employee Arrived Late:");
20 scanf("%d",&late_arrival);
21 //calculations
22 float tot_tax=(tax_rate*gross_salary)/100;
23

```

The console output shows the program's execution with user input and calculated results:

```

D:\que3
enter Employee ID :12
enter Employee Name :xyz
enter Gross salary of the employee :75000
enter Income tax rate in percent :7
enter total leave of the employee in a month :5
enter Number of Allowed Leaves per Month :3
enter Total Number of Days the Employee Arrived Late:6
Emp id   emp name   gross salary   tax rate   tot tax   late arrival   tot leave   net salary
-----
12      xyz      75000.00      7      5250.00      6      5      57250.00
D:\>

```

Question #04

Code:

```
#include<stdio.h>
```

```
int main(){
```

```
    int choice;
```

```
    printf("\nLibrary Management System\n");
```



```
printf("1. Enter book record\n");
printf("2. Enter student record\n");
printf("3. Issue book to a student\n");
printf("4. Return a book and calculate fine\n");
```

```
    printf("Enter your choice: ");
scanf("%d", &choice);
```

```
    if(choice==1){
        //inputs
        char book_id[20];
        char title[50];
        char author_name[20];
        char edition[20];
        char ISBN_number[20];
```

```
        printf("enter id : ");
        scanf("%s",&book_id);
```

```
        printf("enter book title : ");
        scanf("%s",&title);
```

```
        printf("enter author name : ");
        scanf("%s",&author_name);
```

```
        printf("enter edition : ");
        scanf("%s",&edition);
```

```

printf("enter ISBN number : ");
scanf("%s",&ISBN_number);

//outputs
printf("Book id\t title\t author name\t edition\t ISBN number\n");
printf("-----\n");

printf("%s\t %s\t %s\t %s\t %s\t\n",book_id,title,author_name,edition,ISBN_number);
}

```

```

else if(choice==2){

    //inputs
    char std_Reg[20];
    char std_name[20];
    char department[20];
    char section[10];
    char batch[10];

    printf("enter student registration number : ");
    scanf("%s",&std_Reg);

    printf("enter student name : ");
    scanf("%s",&std_name);

    printf("enter department : ");
    scanf("%s",&department);

```

```

printf("enter section : ");
scanf("%s",&section);

printf("enter batch : ");
scanf("%s",&batch);

//outputs
printf("\nregistration number\t student name\t department\t section\t batch\n");
printf("-----\n");
printf("%s#\t %s\t %s\t %s\t %s\t ",std_Reg,std_name,department,section,batch);
}

```

```

else if(choice==3){
    //inputs
    char std_Reg[20];
    char book_id[20];
    char issue_date[20];

    printf("enter student registration number# : ");
    scanf("%s",&std_Reg);

    printf("enter book id : ");
    scanf("%s",&book_id);

    printf("enter issue date : ");
    scanf("%s",&issue_date);

    //outputs

```

```

printf("\nstd.red#\t book id\t issue date\n");
printf("-----\n");
printf("%s\t %s\t %s\t ",std_Reg,book_id,issue_date);
}

```

```

else if(choice==4){

```

```

    //inputs

```

```

    char std_Reg[20];

```

```

    char book_id[20];

```

```

    char issue_date[20];

```

```

    char return_date[20];

```

```

    int fine;

```

```

    printf("enter student registration number# : ");

```

```

    scanf("%s",&std_Reg);

```

```

    printf("enter book id : ");

```

```

    scanf("%s",&book_id);

```

```

    printf("enter issue date : ");

```

```

    scanf("%s",&issue_date);

```

```

    printf("enter return_date : ");

```

```

    scanf("%s",&return_date);

```

```

    printf("enter fine : ");

```

```

    scanf("%d",&fine);

```

```

    //outputs

```

```

    printf("\nstd.red#\t book id\t issue date\t return date\t fine\n");

```

```

    printf("-----\n");

```

```

        printf("%s\t %s\t %s\t %s\t %d\t",std_Reg,book_id,issue_date,return_date,fine);
    }
}

```

ScreenShot:

```

Library Management System
1. Enter book record
2. Enter student record
3. Issue book to a student
4. Return a book and calculate fine
Enter your choice: 3
enter student registration number# : CS-101
enter book id : com-101
enter issue date : 19-12-2023

std.red#          book id          issue date
-----
CS-101    com-101          19-12-2023
-----
Process exited after 35.98 seconds with return value 0
Press any key to continue . . .

```

```

Library Management System
1. Enter book record
2. Enter student record
3. Issue book to a student
4. Return a book and calculate fine
Enter your choice: 1
enter id : Com101
enter book title : ps
enter author name : R.henry
enter edition : 8th
enter ISBN number : 1-292-09881-3
Book id  title  author name  edition  ISBN number
-----
Com101    ps      R.henry      8th      1-292-09881-3
-----
Process exited after 85.43 seconds with return value 0
Press any key to continue . . . |

```

```

Library Management System
1. Enter book record
2. Enter student record
3. Issue book to a student
4. Return a book and calculate fine
Enter your choice: 4
enter student registration number# : CS-101
enter book id : com-101
enter issue date : 1-12-2023
enter return_date : 30-12-2023
enter fine : 450

std.red#          book id          issue date          return date          fine
-----
CS-101    com-101          1-12-2023          30-12-2023          450
-----
Process exited after 52.22 seconds with return value 0
Press any key to continue . . .

```

```

Library Management System
1. Enter book record
2. Enter student record
3. Issue book to a student
4. Return a book and calculate fine
Enter your choice: 2
enter student registration number : CS-101
enter student name : Ali
enter department : CS
enter section : B05
enter batch : Fall-2023

registration number    student name    department    section    batch
-----
CS-101    Ali    CS    B05    Fall-2023
-----
Process exited after 44.08 seconds with return value 0
Press any key to continue . . . |

```

Question #05

Code:

```
#include<stdio.h>
```

```
int main(){
```

```
//part a
```

```
int num=17;
```

```
num=(num++%3)+(--num*2);
```

```
printf("The final value of num is %d",num);
```

```
//part b
```

```
int a=5;
```

```
int b=6;
```

```
if(++a==b--){
```

```
    sum = a+b
```

```
    printf("if statement executed");
```

```
}
```

```
//part c
```

```
int base=2;
```

```
int exponent=3;
```

```
int result=1;
```

```
result*=base++;
```

```
printf("The final value of base is %d",result);
```

```
//part d
```

```
int a=3;
```

```
int b=++a+a++ + --a;
```

```
printf("Value of b is %d",b);
```

```
//part e
```

```
int x=2,y=2;
```

```
x/=x/y;
```

```
printf("%d\n",x);
```

```
//part f
```

A is the correct answer

```
}
```

ScreenShot:

The screenshot displays the Embarcadero Dev-C++ 6.3 IDE. The main editor window shows a C program with the following code:

```
1 #include<stdio.h>
2 int main(){
3
4 //part a
5
6 int num=17;
7 num=(num++%3)+(--num*2);
8 printf("The final value of num is %d",num);
9
10
11 //part b
12 int a=5;
13 int b=6;
14 if(++a==b--){
15     sum = a+b
```

The console window at the bottom shows the execution output:

```
PS D:\que5
a=b
$1 D:\>gcc -o que5 que5.c
$2 D:\>que5
The final value of base is 2 :
$3 D:\>gcc -o que5 que5.c
D:\>que5
Value of b is 13
D:\>gcc -o que5 que5.c
D:\>que5
2
```

The status bar at the bottom indicates the current line is 37, column is 37, and the file is 513 lines long.